
Program 1. : Program to Create the equivalent of a four-function calculator. The Program requires the user to enter two numbers and an operator. It then carries out the specified arithmetical operation: addition, subtraction, multiplication or division of the two numbers. Finally, it displays the result.

Soln:

```
#include <iostream>

using namespace std;

int main() {

    char choice;

    do {

        double num1, num2, result;

        char operators;

        cout << "Enter first number: ";

        cin >> num1;

        cout << "Enter an operator (+, -, *, /): ";

        cin >> operators;

        cout << "Enter second number: ";

        cin >> num2;

        if (operators == '+') {

            result = num1 + num2;

            cout << "Result: " << result << endl;

        } else if (operators == '-') {

            result = num1 - num2;

            cout << "Result: " << result << endl;

        } else if (operators == '*') {

            result = num1 * num2;

            cout << "Result: " << result << endl;

        } else if (operators == '/') {

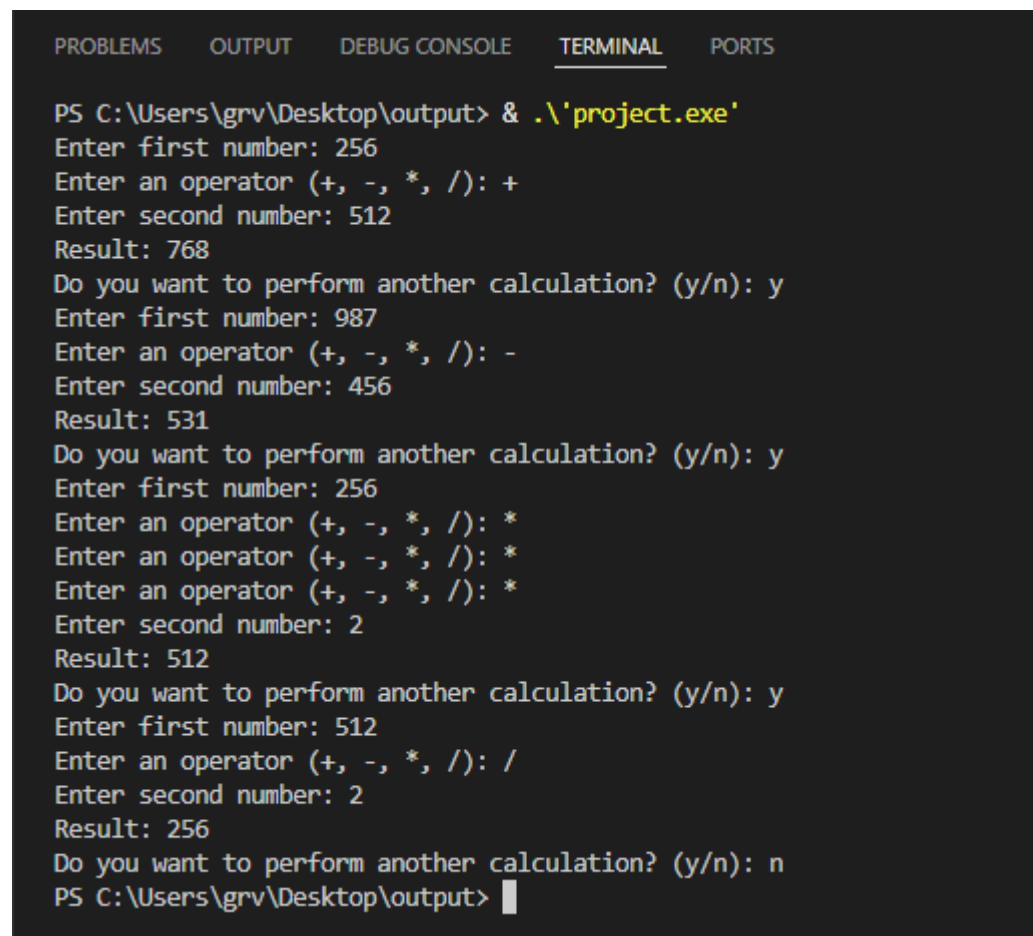
            if (num2 != 0) {

                result = num1 / num2;

                cout << "Result: " << result << endl;

            } else {
```

```
        cout << "Division by zero is not allowed." << endl;
    }
} else {
    cout << "Invalid operator!" << endl;
}
cout << "Do you want to perform another calculation? (y/n): ";
cin >> choice;
} while (choice == 'y' || choice == 'Y');
return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\grv\Desktop\output> & .\'project.exe'
Enter first number: 256
Enter an operator (+, -, *, /): +
Enter second number: 512
Result: 768
Do you want to perform another calculation? (y/n): y
Enter first number: 987
Enter an operator (+, -, *, /): -
Enter second number: 456
Result: 531
Do you want to perform another calculation? (y/n): y
Enter first number: 256
Enter an operator (+, -, *, /): *
Enter an operator (+, -, *, /): *
Enter an operator (+, -, *, /): *
Enter second number: 2
Result: 512
Do you want to perform another calculation? (y/n): y
Enter first number: 512
Enter an operator (+, -, *, /): /
Enter second number: 2
Result: 256
Do you want to perform another calculation? (y/n): n
PS C:\Users\grv\Desktop\output> █
```

Program 2. : Program to input a character and to print whether a given character is an alphabet, digit or any other character.

Soln:

```
#include <iostream>

using namespace std;

int main() {

    char inputChar, userChoice;

    do {

        cout << "Enter a character: ";

        cin >> inputChar;

        if ((inputChar >= 'A' && inputChar <= 'Z') || (inputChar >= 'a' && inputChar <= 'z')) {

            cout << "The character '" << inputChar << "' is an alphabet." << endl;

        } else if (inputChar >= '0' && inputChar <= '9') {

            cout << "The character '" << inputChar << "' is a digit." << endl;

        } else {

            cout << "The character '" << inputChar << "' is a special character." << endl;

        }

        cout << "Do you want to check another character? (y/n): ";

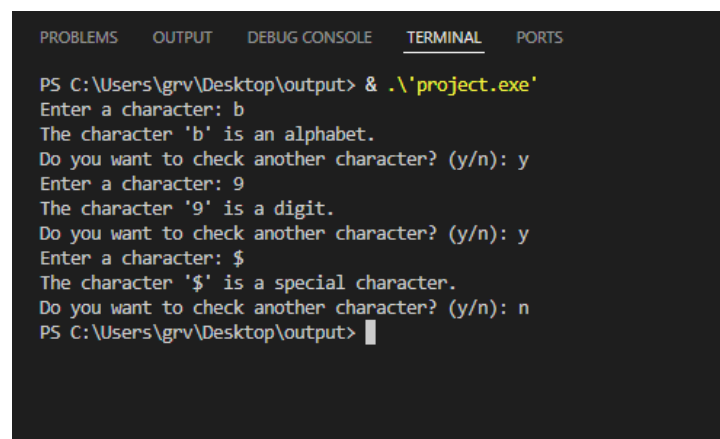
        cin >> userChoice;

    } while (userChoice == 'y' || userChoice == 'Y');

    return 0;

}
```

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\grv\Desktop\output> & .\project.exe
Enter a character: b
The character 'b' is an alphabet.
Do you want to check another character? (y/n): y
Enter a character: 9
The character '9' is a digit.
Do you want to check another character? (y/n): y
Enter a character: $
The character '$' is a special character.
Do you want to check another character? (y/n): n
PS C:\Users\grv\Desktop\output> 
```

Program 3. : Program to calculate and print roots of a quadratic equation $ax^2+bx+c = 0$ ($a \neq 0$).

Soln:

```
#include <iostream>

#include <cmath>

using namespace std;

int main() {

    char userChoice;

    do {

        double a, b, c;

        double discriminant, root1, root2;

        cout << "Enter a: ";

        cin >> a;

        while (a == 0) {

            cout << "a cannot be zero. Enter again: ";

            cin >> a;

        }

        cout << "Enter b: ";

        cin >> b;

        cout << "Enter c: ";

        cin >> c;

        discriminant = b*b - 4*a*c;

        if (discriminant > 0) {

            root1 = (-b + sqrt(discriminant)) / (2*a);

            root2 = (-b - sqrt(discriminant)) / (2*a);

            cout << "Roots are real and distinct" << endl;

            cout << root1 << " " << root2 << endl;

        } else if (discriminant == 0) {

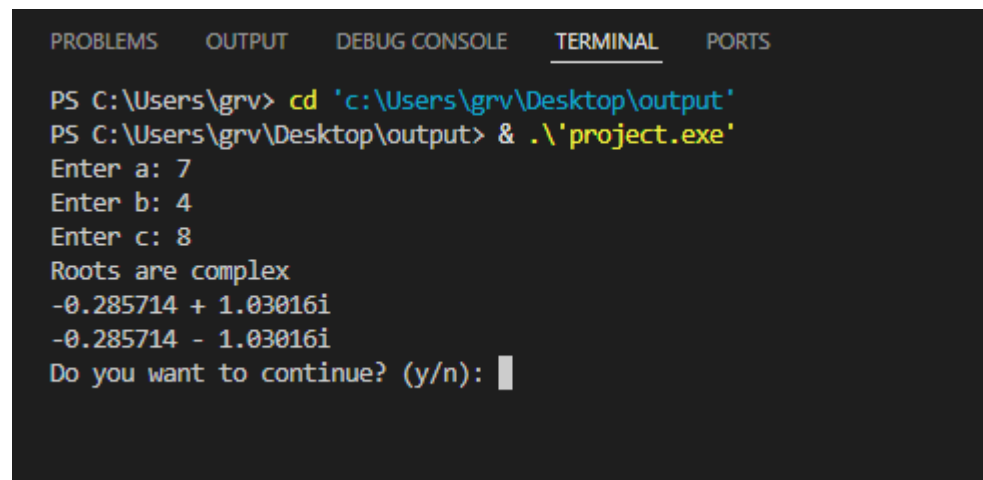
            root1 = -b / (2*a);

            cout << "Roots are real and equal: " << root1 << endl;

        } else {

            double real = -b / (2*a);
```

```
        double imag = sqrt(-discriminant) / (2*a);  
        cout << "Roots are complex" << endl;  
        cout << real << " + " << imag << "i" << endl;  
        cout << real << " - " << imag << "i" << endl;  
    }  
    cout << "Do you want to continue? (y/n): ";  
    cin >> userChoice;  
} while (userChoice == 'y' || userChoice == 'Y');  
return 0;  
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
PS C:\Users\grv> cd 'c:\Users\grv\Desktop\output'  
PS C:\Users\grv\Desktop\output> & .\project.exe  
Enter a: 7  
Enter b: 4  
Enter c: 8  
Roots are complex  
-0.285714 + 1.03016i  
-0.285714 - 1.03016i  
Do you want to continue? (y/n):
```

Program 4. : Program to calculate area of a circle, a rectangle or a triangle depending upon user's choice.

Soln:

```
#include <iostream>

#include <cmath>

using namespace std;

int main(){
    int choice;
    char repeat;
    do{
        cout << "\nChoose the shape to calculate the area:" << endl;
        cout << "1. Circle" << endl;
        cout << "2. Rectangle" << endl;
        cout << "3. Triangle" << endl;
        cout << "Enter your choice (1-3): ";
        cin >> choice;
        switch (choice){
            case 1:
            {
                double radius, area;
                cout << "Enter the radius of the circle: ";
                cin >> radius;
                if (radius < 0) {
                    cout << "Radius cannot be negative!" << endl;
                }else{
                    area = M_PI * radius * radius;
                    cout << "The area of the circle is: " << area << endl;
                }
                break;
            }
            case 2:
```

```
{  
    double length, width, area;  
    cout << "Enter the length of the rectangle: ";  
    cin >> length;  
    cout << "Enter the width of the rectangle: ";  
    cin >> width;  
    if (length < 0 || width < 0){  
        cout << "Length and width cannot be negative!" << endl;  
    }else{  
        area = length * width;  
        cout << "The area of the rectangle is: " << area << endl;  
    }  
    break;  
}  
case 3:  
{  
    double base, height, area;  
    cout << "Enter the base of the triangle: ";  
    cin >> base;  
    cout << "Enter the height of the triangle: ";  
    cin >> height;  
    if (base < 0 || height < 0) {  
        cout << "Base and height cannot be negative!" << endl;  
    }else{  
        area = 0.5 * base * height;  
        cout << "The area of the triangle is: " << area << endl;  
    }  
    break;  
}  
default:
```

```
        cout << "Invalid choice! Please choose a valid option (1-3)." << endl;
    }

    cout << "\nDo you want to calculate the area of another shape? (y/n): ";

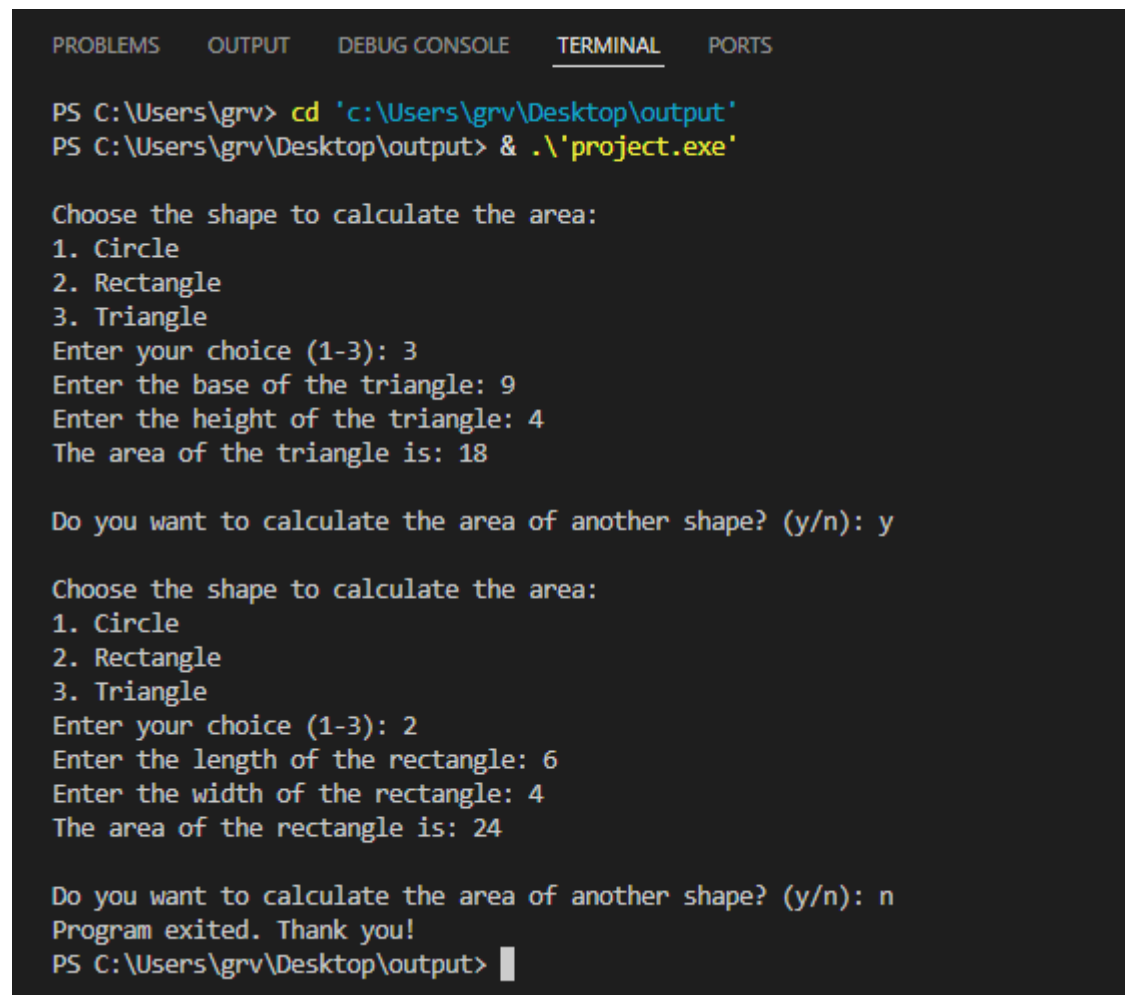
    cin >> repeat;

} while (repeat == 'y' || repeat == 'Y');

cout << "Program exited. Thank you!" << endl;

return 0;

}
```

Output:A screenshot of a code editor's terminal window showing the execution of a C++ program. The terminal has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (selected), and PORTS. The command prompt shows the user navigating to 'c:\Users\grv\Desktop\output' and running 'project.exe'. The program prompts the user to choose a shape (1-3) and calculate its area. In the first run, the user chooses a triangle (3), enters a base of 9 and a height of 4, and the program outputs an area of 18. The user then chooses 'y' to calculate another shape. In the second run, the user chooses a rectangle (2), enters a length of 6 and a width of 4, and the program outputs an area of 24. Finally, the user chooses 'n' to exit, and the program outputs 'Program exited. Thank you!' before returning to the command prompt.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\grv> cd 'c:\Users\grv\Desktop\output'
PS C:\Users\grv\Desktop\output> & .\'project.exe'

Choose the shape to calculate the area:
1. Circle
2. Rectangle
3. Triangle
Enter your choice (1-3): 3
Enter the base of the triangle: 9
Enter the height of the triangle: 4
The area of the triangle is: 18

Do you want to calculate the area of another shape? (y/n): y

Choose the shape to calculate the area:
1. Circle
2. Rectangle
3. Triangle
Enter your choice (1-3): 2
Enter the length of the rectangle: 6
Enter the width of the rectangle: 4
The area of the rectangle is: 24

Do you want to calculate the area of another shape? (y/n): n
Program exited. Thank you!
PS C:\Users\grv\Desktop\output> |
```

Program 5. : Program to calculate the factorial of a positive integer.

Soln:

```
#include <iostream>

using namespace std;

int main(){

    int n;

    char ch;

    do{

        cout << "Enter a positive integer to calculate its factorial: ";

        cin >> n;

        unsigned long long fact = 1;

        for (int i = 1; i <= n; i++){

            fact *= i;

        }

        cout << "Factorial = " << fact << endl;

        cout << "Repeat? (y/n): ";

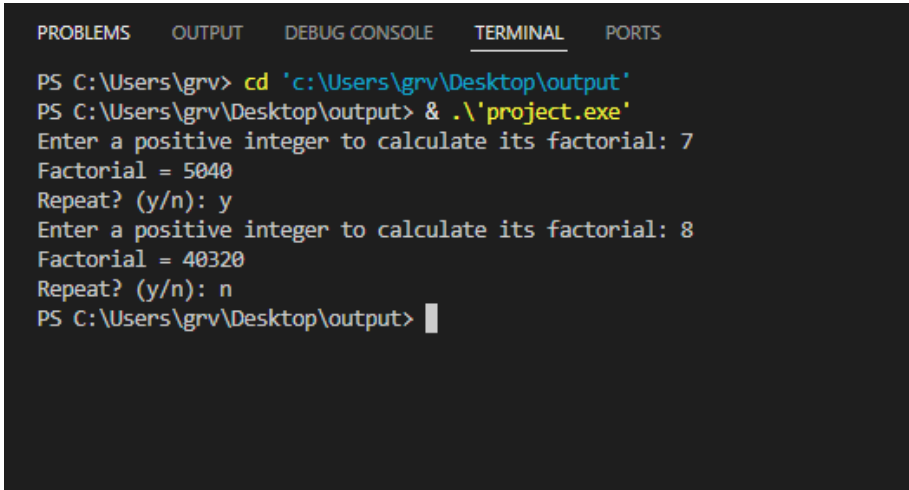
        cin >> ch;

    } while (ch == 'y' || ch == 'Y');

    return 0;

}
```

Output :



The screenshot shows a terminal window with tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active. The prompt is PS C:\Users\grv>. The user enters 'cd 'c:\Users\grv\Desktop\output'', followed by '& .\'project.exe\''. The program prompts 'Enter a positive integer to calculate its factorial: 7', outputs 'Factorial = 5040', and asks 'Repeat? (y/n): y'. The user enters 'y'. The program prompts 'Enter a positive integer to calculate its factorial: 8', outputs 'Factorial = 40320', and asks 'Repeat? (y/n): n'. The user enters 'n'. The prompt returns to PS C:\Users\grv\Desktop\output>.

```
PS C:\Users\grv> cd 'c:\Users\grv\Desktop\output'
PS C:\Users\grv\Desktop\output> & .\'project.exe'
Enter a positive integer to calculate its factorial: 7
Factorial = 5040
Repeat? (y/n): y
Enter a positive integer to calculate its factorial: 8
Factorial = 40320
Repeat? (y/n): n
PS C:\Users\grv\Desktop\output>
```

Program 6. : Program to search for a specific integer in a 1-D array (Linear Search).

Soln:

```
#include <iostream>

using namespace std;

int main(){

    int size, target, found = 0;

    char userChoice;

    do{

        cout << "Enter the size of the array (positive integer): ";

        cin >> size;

        while (size <= 0){

            cout << "Array size must be a positive integer. Please enter again: ";

            cin >> size;

        }

        int array[size];

        cout << "Enter " << size << " integers for the array: " << endl;

        for (int i = 0; i < size; i++){

            cout << "Element " << i + 1 << ": ";

            cin >> array[i];

        }

        cout << "Enter the integer to search in the array: ";

        cin >> target;

        found = 0;

        for (int i = 0; i < size; i++){

            if (array[i] == target){

                cout << "The integer " << target << " is found at position " << i + 1 << " (index " << i << ")."

                << endl;

                found = 1;

                break;

            }

        }

        if (!found)

        {
```

```
    cout << "The integer " << target << " is not present in the array." << endl;
}

    cout << "Do you want to search in another array? (y/n): ";

    cin >> userChoice;

} while (userChoice == 'y' || userChoice == 'Y');

    cout << "Program exited. Thank you!" << endl;

    return 0;

}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\grv> cd 'c:\Users\grv\Desktop\output'
PS C:\Users\grv\Desktop\output> & .\'project.exe'
Enter the size of the array (positive integer): 5
Enter 5 integers for the array:
Element 1: 3
Element 2: 5
Element 3: 7
Element 4: 3
Element 5: 5
Enter the integer to search in the array: 7
The integer 7 is found at position 3 (index 2).
Do you want to search in another array? (y/n): n
Program exited. Thank you!
PS C:\Users\grv\Desktop\output> █
```

Program 7. : Program to count the number of employees earning more than Rs 1 lakh per annum. The monthly salaries of 20 employees shall be provided by user.

Soln:

```
#include <iostream>

using namespace std;

int main(){

    const int numEmployees = 20;

    double monthlySalaries[numEmployees];

    int count = 0;

    cout << "Enter the monthly salaries of " << numEmployees << " employees (in ₹):" << endl;

    for (int i = 0; i < numEmployees; i++){

        cout << "Employee " << i + 1 << ": ";

        cin >> monthlySalaries[i];

        while (monthlySalaries[i] < 0){

            cout << "Monthly salary cannot be negative. Please re-enter: ";

            cin >> monthlySalaries[i];

        }

    }

    for (int i = 0; i < numEmployees; i++){

        double annualSalary = monthlySalaries[i] * 12;

        if (annualSalary > 100000){

            count++;

        }

    }

    cout << "\nThe number of employees earning more than ₹1 lakh per annumis: " << count << endl;

    return 0;

}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\grv\Desktop\output> & .\project.exe'
Enter the monthly salaries of 20 employees (in ₹):
Employee 1: 9999
Employee 2: 546763
Employee 3: 68593
Employee 4: 8475334
Employee 5: 23284
Employee 6: 68000
Employee 7: 72999
Employee 8: 843843
Employee 9: 8374839
Employee 10: 9889993
Employee 11: 932543
Employee 12: 4376437
Employee 13: 456654
Employee 14: 3546577
Employee 15: 456543
Employee 16: 987766
Employee 17: 4564
Employee 18: 543
Employee 19: 456744
Employee 20: 365754

The number of employees earning more than ₹1 lakh per annumis: 18
PS C:\Users\grv\Desktop\output> 
```

Program 8. : Program to add two matrices.**Soln:**

```
#include <iostream>

using namespace std;

int main(){

    int rows, cols;

    cout << "Enter the number of rows for the matrices: ";

    cin >> rows;

    cout << "Enter the number of columns for the matrices: ";

    cin >> cols;

    int matrixA[rows][cols], matrixB[rows][cols], result[rows][cols];

    cout << "\nEnter elements of the first matrix (Matrix A):" << endl;

    for (int i = 0; i < rows; i++){

        for (int j = 0; j < cols; j++){

            cout << "Element [" << i + 1 << "][" << j + 1 << "]: ";

            cin >> matrixA[i][j];

        }

    }

    cout << "\nEnter elements of the second matrix (Matrix B):" << endl;

    for (int i = 0; i < rows; i++){

        for (int j = 0; j < cols; j++){

            cout << "Element [" << i + 1 << "][" << j + 1 << "]: ";

            cin >> matrixB[i][j];

        }

    }

    for (int i = 0; i < rows; i++){

        for (int j = 0; j < cols; j++){

            result[i][j] = matrixA[i][j] + matrixB[i][j];

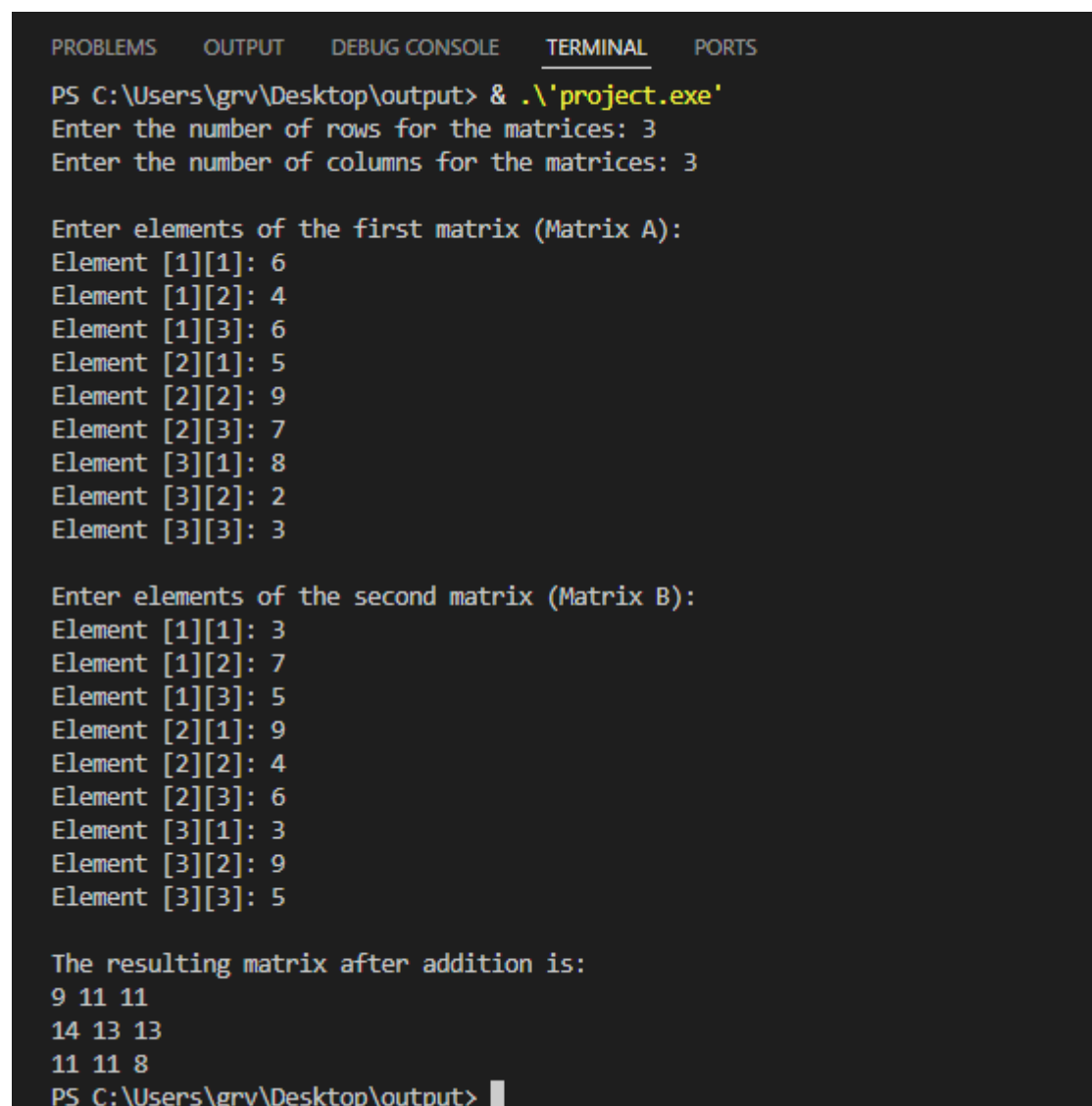
        }

    }

    cout << "\nThe resulting matrix after addition is:" << endl;

    for (int i = 0; i < rows; i++){
```

```
    for (int j = 0; j < cols; j++)  
    {  
        cout << result[i][j] << " ";  
    }  
    cout << endl;  
}  
return 0;  
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
PS C:\Users\grv\Desktop\output> & .\'project.exe'  
Enter the number of rows for the matrices: 3  
Enter the number of columns for the matrices: 3  
  
Enter elements of the first matrix (Matrix A):  
Element [1][1]: 6  
Element [1][2]: 4  
Element [1][3]: 6  
Element [2][1]: 5  
Element [2][2]: 9  
Element [2][3]: 7  
Element [3][1]: 8  
Element [3][2]: 2  
Element [3][3]: 3  
  
Enter elements of the second matrix (Matrix B):  
Element [1][1]: 3  
Element [1][2]: 7  
Element [1][3]: 5  
Element [2][1]: 9  
Element [2][2]: 4  
Element [2][3]: 6  
Element [3][1]: 3  
Element [3][2]: 9  
Element [3][3]: 5  
  
The resulting matrix after addition is:  
9 11 11  
14 13 13  
11 11 8  
PS C:\Users\grv\Desktop\output>
```

Program 9. : Program to read Sales of 5 salesman in 12 months and to print total sales made by each salesman.

Soln:

```
#include <iostream>

using namespace std;

int main(){

    const int salesmen = 5;

    const int months = 12;

    double sales[salesmen][months];

    double totalSales[salesmen] = {0};

    cout << "Enter the sales data for " << salesmen << " salesmen for " << months << "
months:\n";

    for (int i = 0; i < salesmen; i++){

        cout << "\nSalesman " << i + 1 << ": " << endl;

        for (int j = 0; j < months; j++){

            cout << "Month " << j + 1 << ": rs";

            cin >> sales[i][j];

            while (sales[i][j] < 0)

            {

                cout << "Sales cannot be negative. Please re-enter for Month " << j + 1 << ": ₹";

                cin >> sales[i][j];

            }

            totalSales[i] += sales[i][j];

        }

    }

    cout << "\nTotal sales made by each salesman:\n";

    for (int i = 0; i < salesmen; i++){

        cout << "Salesman " << i + 1 << ": rs" << totalSales[i] << endl;

    }

    return 0;

}
```


Output:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\grv\Desktop\output> & .\project.exe
Enter the sales data for 5 salesmen for 12 months:

Salesman 1:      Salesman 3:      Salesman 4:
Month 1: rs999   Month 1: rs5684   Month 1: rs2364
Month 2: rs999   Month 2: rs3475643 Month 2: rs45636
Month 3: rs999   Month 3: rs347    Month 3: rs4356
Month 4: rs999   Month 4: rs34573  Month 4: rs3456
Month 5: rs999   Month 5: rs73657  Month 5: rs64337
Month 6: rs9999  Month 6: rs8657   Month 6: rs56383
Month 7: rs999   Month 7: rs569    Month 7: rs47845
Month 8: rs876   Month 8: rs5678   Month 8: rs86595
Month 9: rs565   Month 9: rs5687   Month 9: rs5694
Month 10: rs758  Month 10: rs468   Month 10: rs45864
Month 11: rs634  Month 11: rs456   Month 11: rs86479
Month 12: rs754  Month 12: rs574   Month 12: rs48446

Salesman 2:      Salesman 5:
Month 1: rs23456 Month 1: rs568769
Month 2: rs6543  Month 2: rs9676
Month 3: rs54643 Month 3: rs444
Month 4: rs36437 Month 4: rs64576
Month 5: rs43564 Month 5: rs58645
Month 6: rs3456345 Month 6: rs548
Month 7: rs8785  Month 7: rs4848
Month 8: rs65785 Month 8: rs4784
Month 9: rs576   Month 9: rs4674
Month 10: rs8756 Month 10: rs846
Month 11: rs45684 Month 11: rs467848
Month 12: rs4637 Month 12: rs978

Total sales made by each salesman:
Salesman 1: rs19580
Salesman 2: rs3.75521e+006
Salesman 3: rs3.61199e+006
Salesman 4: rs497455
Salesman 5: rs1.18664e+006
PS C:\Users\grv\Desktop\output>

```

Program 10. : Program to calculate grades of 4 students from 3 test scores.

Soln:

```
#include <iostream>

using namespace std;

int main(){

    int num_students = 4;

    int num_tests = 3;

    float scores[num_students][num_tests];

    float total, average;

    for (int i = 0; i < num_students; i++){

        total = 0;

        cout << "\nEnter scores for Student " << i + 1 << ":\n";

        for (int j = 0; j < num_tests; j++){

            cout << "Enter score for Test " << j + 1 << " ";

            cin >> scores[i][j];

            total += scores[i][j];

        }

        average = total / num_tests;

        cout << "Student " << i + 1 << " - Total: " << total << ", Average: " << average << ", Grade: ";

        if (average >= 90){

            cout << "A\n";

        }else if (average >= 80){

            cout << "B\n";

        }else if (average >= 70){

            cout << "C\n";

        }else if (average >= 60){

            cout << "D\n";

        }else

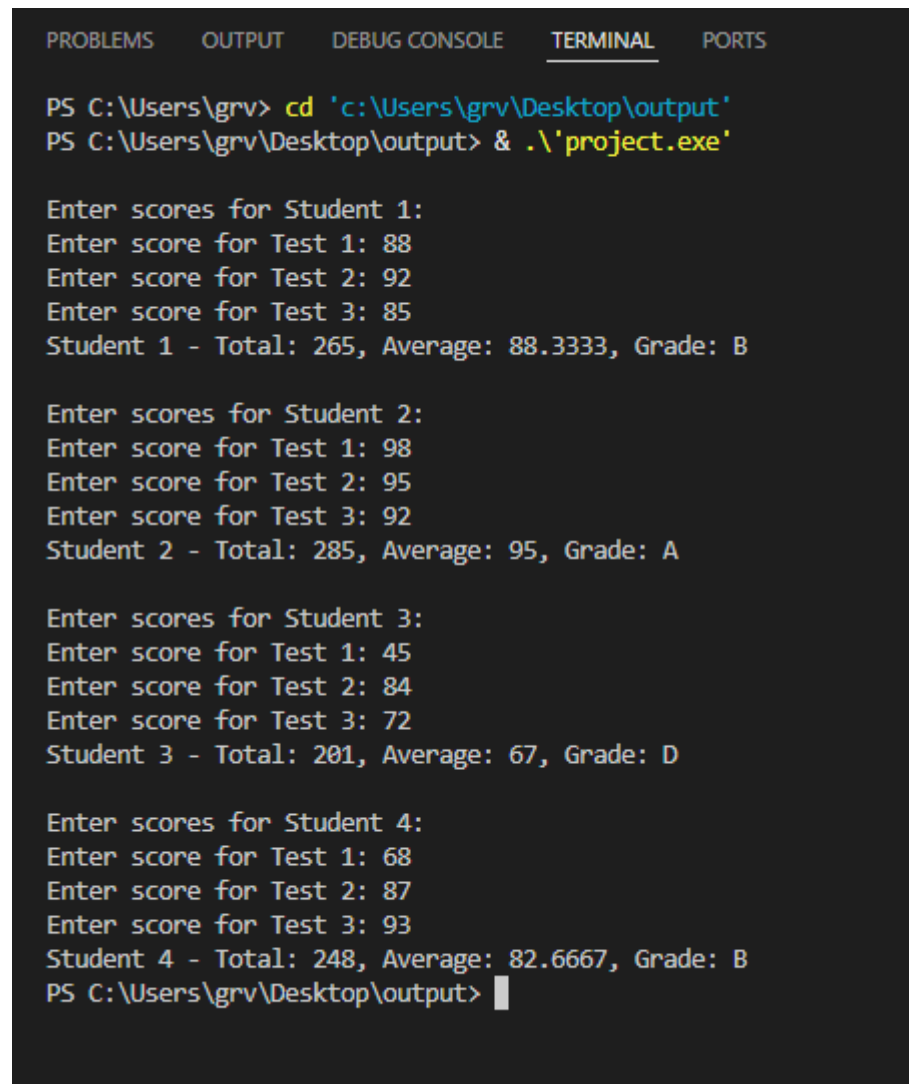
        {

            cout << "F\n";

        }

    }
```

```
}  
  
}  
  
return 0;  
  
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
  
PS C:\Users\grv> cd 'c:\Users\grv\Desktop\output'  
PS C:\Users\grv\Desktop\output> & .\'project.exe'  
  
Enter scores for Student 1:  
Enter score for Test 1: 88  
Enter score for Test 2: 92  
Enter score for Test 3: 85  
Student 1 - Total: 265, Average: 88.3333, Grade: B  
  
Enter scores for Student 2:  
Enter score for Test 1: 98  
Enter score for Test 2: 95  
Enter score for Test 3: 92  
Student 2 - Total: 285, Average: 95, Grade: A  
  
Enter scores for Student 3:  
Enter score for Test 1: 45  
Enter score for Test 2: 84  
Enter score for Test 3: 72  
Student 3 - Total: 201, Average: 67, Grade: D  
  
Enter scores for Student 4:  
Enter score for Test 1: 68  
Enter score for Test 2: 87  
Enter score for Test 3: 93  
Student 4 - Total: 248, Average: 82.6667, Grade: B  
PS C:\Users\grv\Desktop\output> |
```

Program 11. : Write a program that takes a string as input and outputs the reversed string. For example, if the input is "Skillarger", the output should be "regallikS".

Soln:

```
#include <iostream>

#include <string>

using namespace std;

int main(){

    string originalString, reversedString = "";

    cout << "Enter a string: ";

    cin >> originalString;

    int stringLength = originalString.length();

    for (int index = stringLength - 1; index >= 0; index--){

        reversedString += originalString[index];

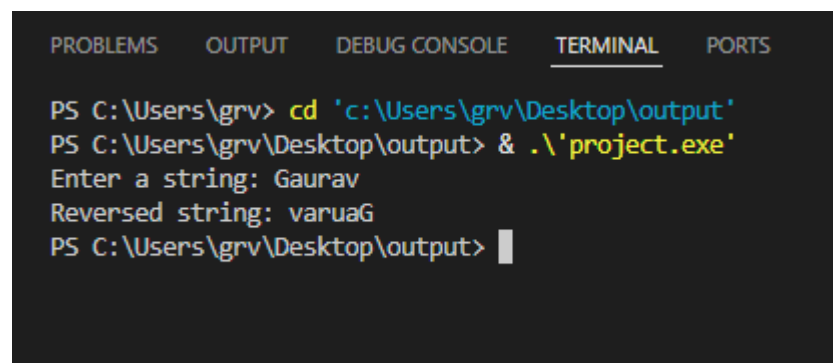
    }

    cout << "Reversed string: " << reversedString << endl;

    return 0;

}
```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\grv> cd 'c:\Users\grv\Desktop\output'
PS C:\Users\grv\Desktop\output> & .\'project.exe'
Enter a string: Gaurav
Reversed string: varuaG
PS C:\Users\grv\Desktop\output> |
```

Program 12. : Write a program to count the number of vowels and consonants in a given string. For example, for the input "Computer", the output should be Vowels: 3, Consonants: 5.

Soln:

```
#include <iostream>

#include <string>

using namespace std;

int main(){

    string inputString;

    int vowelsCount = 0, consonantsCount = 0;

    cout << "Enter a string: ";

    cin >> inputString;

    for (int i = 0; i < inputString.length(); i++){

        char ch = tolower(inputString[i]);

        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u'){

            vowelsCount++;

        }else if ((ch >= 'a' && ch <= 'z'))

        {

            consonantsCount++;

        }

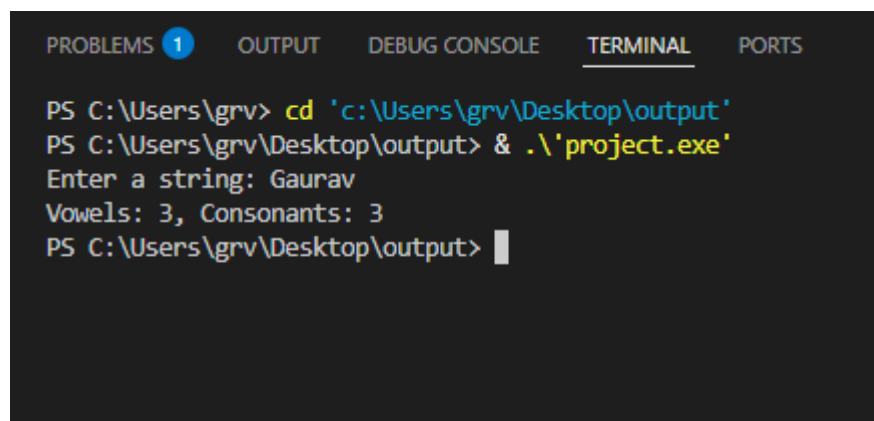
    }

    cout << "Vowels: " << vowelsCount << ", Consonants: " << consonantsCount << endl;

    return 0;

}
```

Output:



```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\grv> cd 'c:\Users\grv\Desktop\output'
PS C:\Users\grv\Desktop\output> & .\project.exe
Enter a string: Gaurav
Vowels: 3, Consonants: 3
PS C:\Users\grv\Desktop\output> |
```

Program 13. : Write a program to check whether a given string is a palindrome. For example, "level" is a palindrome, but "hello" is not.

Soln:

```
#include <iostream>

#include <string>

using namespace std;

int main(){

    string inputString;

    bool isPalindrome = true;

    cout << "Enter a string: ";

    cin >> inputString;

    int start = 0;

    int end = inputString.length() - 1;

    while (start < end){

        if (tolower(inputString[start]) != tolower(inputString[end])){

            isPalindrome = false;

            break;

        }

        start++;

        end--;

    }

    if (isPalindrome){

        cout << "The string is a palindrome." << endl;

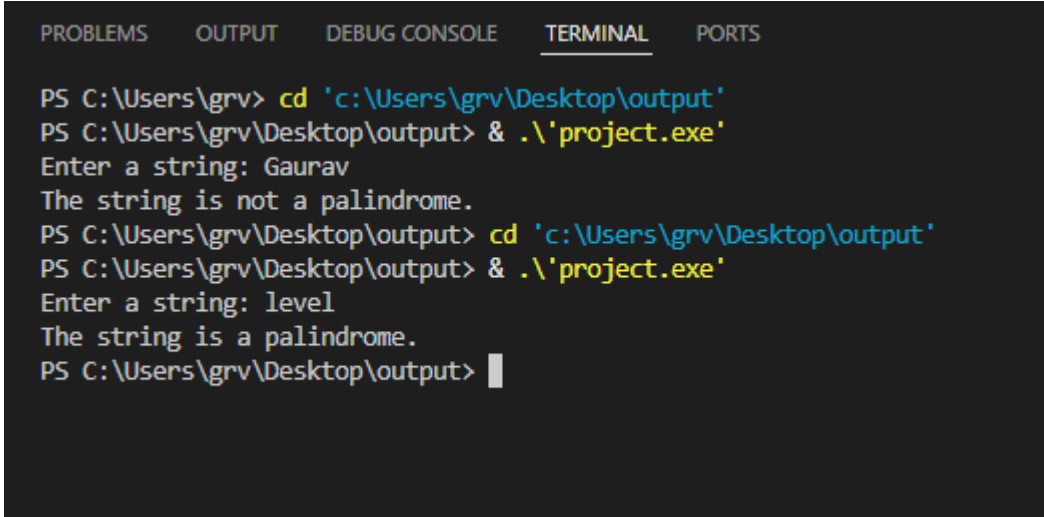
    }else{

        cout << "The string is not a palindrome." << endl;

    }

    return 0;

}
```

Output:

The screenshot shows a terminal window with a dark background and light-colored text. At the top, there are tabs labeled 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. The terminal content shows a series of commands and outputs: a directory change to 'c:\Users\grv\Desktop\output', running a program named 'project.exe', entering the string 'Gaurav', receiving the output 'The string is not a palindrome.', changing the directory back to 'c:\Users\grv\Desktop\output', running 'project.exe' again, entering the string 'level', and receiving the output 'The string is a palindrome.'. The prompt 'PS C:\Users\grv\Desktop\output>' is visible at the end of the last line.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\grv> cd 'c:\Users\grv\Desktop\output'
PS C:\Users\grv\Desktop\output> & .\'project.exe'
Enter a string: Gaurav
The string is not a palindrome.
PS C:\Users\grv\Desktop\output> cd 'c:\Users\grv\Desktop\output'
PS C:\Users\grv\Desktop\output> & .\'project.exe'
Enter a string: level
The string is a palindrome.
PS C:\Users\grv\Desktop\output> 
```

Program 14. : Write a program that removes duplicate characters from a string. For example, if the input is "programming", the output should be "progamin".

Soln:

```
#include <iostream>

#include <string>

using namespace std;

int main(){

    string inputString, resultString = "";

    cout << "Enter a string: ";

    cin >> inputString;

    for (int i = 0; i < inputString.length(); i++){

        bool isDuplicate = false;

        for (int j = 0; j < resultString.length(); j++){

            if (inputString[i] == resultString[j]){

                isDuplicate = true;

                break;

            }

        }

        if (!isDuplicate){

            resultString += inputString[i];

        }

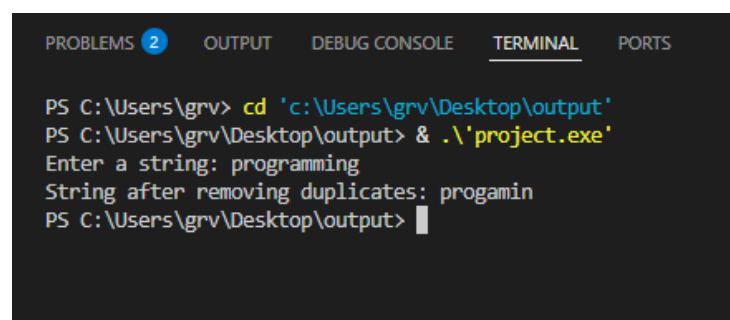
    }

    cout << "String after removing duplicates: " << resultString << endl;

    return 0;

}
```

Output:



```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\grv> cd 'c:\Users\grv\Desktop\output'
PS C:\Users\grv\Desktop\output> & .\project.exe
Enter a string: programming
String after removing duplicates: progamin
PS C:\Users\grv\Desktop\output> 
```

Program 15. : Write a program to find the longest word in a given sentence. For example, for the input "C++ is a powerful programming language", the output should be "programming".

Soln:

```
#include <iostream>

#include <sstream>

#include <string>

using namespace std;

int main(){

    string sentence, word, longestWord = "";

    int maxLength = 0;

    cout << "Enter a sentence: ";

    getline(cin, sentence);

    stringstream ss(sentence);

    while (ss >> word){

        if (word.length() > maxLength){

            longestWord = word;

            maxLength = word.length();

        }

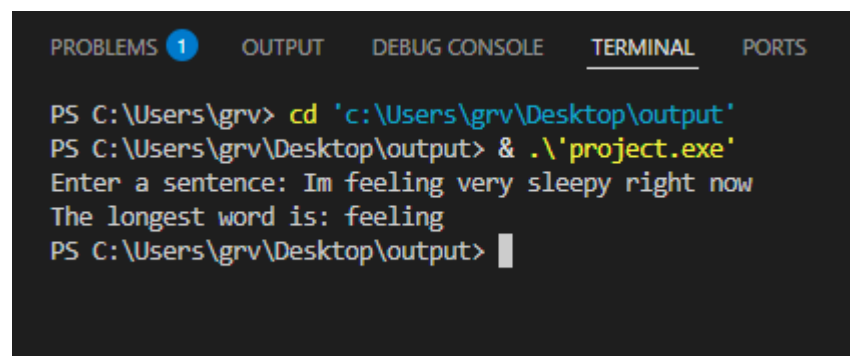
    }

    cout << "The longest word is: " << longestWord << endl;

    return 0;

}
```

Output:



```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\grv> cd 'c:\Users\grv\Desktop\output'
PS C:\Users\grv\Desktop\output> & .\'project.exe'
Enter a sentence: Im feeling very sleepy right now
The longest word is: feeling
PS C:\Users\grv\Desktop\output> |
```

Program 16. : Program to read values into a nested structure.

Soln:

```
#include <iostream>

#include <string>

using namespace std;

struct Student{

    string name;

    int age;

    float marks[3];

};

struct Class{

    string className;

    Student students[3];

};

int main(){

    Class classData;

    cout << "Enter the class name: ";

    getline(cin, classData.className);

    for (int i = 0; i < 2; i++){

        cout << "\nEnter details for Student " << i + 1 << ":\n";

        cout << "Name: ";

        getline(cin, classData.students[i].name);

        cout << "Age: ";

        cin >> classData.students[i].age;

        for (int j = 0; j < 3; j++){

            cout << "Marks for subject " << j + 1 << " : ";

            cin >> classData.students[i].marks[j];

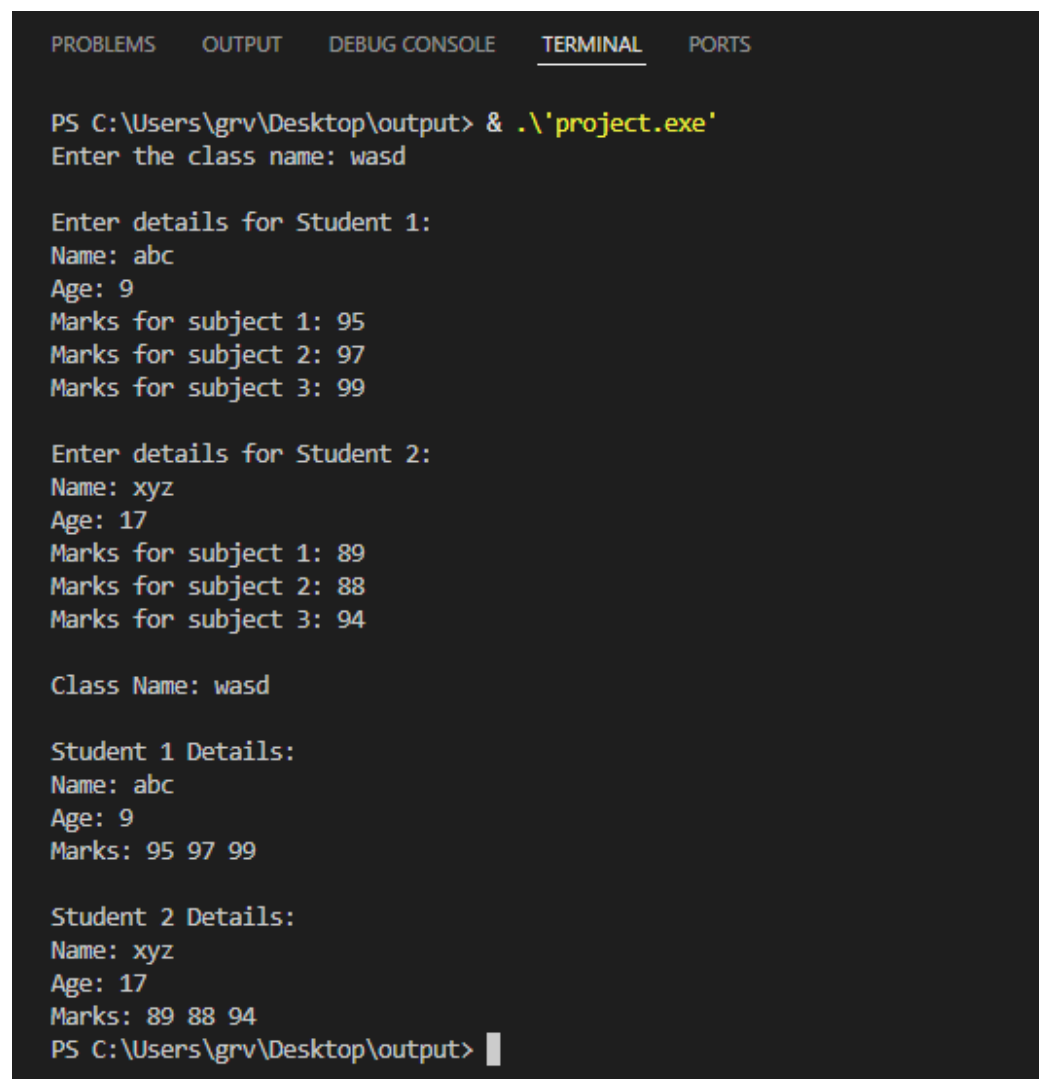
        }

        cin.ignore();

    }

    cout << "\nClass Name: " << classData.className << endl;
```

```
for (int i = 0; i < 2; i++){  
    cout << "\nStudent " << i + 1 << " Details:\n";  
    cout << "Name: " << classData.students[i].name << endl;  
    cout << "Age: " << classData.students[i].age << endl;  
    cout << "Marks: ";  
    for (int j = 0; j < 3; j++){  
        cout << classData.students[i].marks[j] << " ";  
    }  
    cout << endl;  
}  
return 0;  
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
  
PS C:\Users\grv\Desktop\output> & .\'project.exe'  
Enter the class name: wasd  
  
Enter details for Student 1:  
Name: abc  
Age: 9  
Marks for subject 1: 95  
Marks for subject 2: 97  
Marks for subject 3: 99  
  
Enter details for Student 2:  
Name: xyz  
Age: 17  
Marks for subject 1: 89  
Marks for subject 2: 88  
Marks for subject 3: 94  
  
Class Name: wasd  
  
Student 1 Details:  
Name: abc  
Age: 9  
Marks: 95 97 99  
  
Student 2 Details:  
Name: xyz  
Age: 17  
Marks: 89 88 94  
PS C:\Users\grv\Desktop\output> |
```

Program 17. : Program to store information of 10 employees and to display information of an employee depending upon the employee no given.

Soln:

```
#include <iostream>

#include <string>

using namespace std;

struct Employee{
    int employeeNumber;
    string employeeName, employeeDepartment;
    float employeeSalary;
};

void inputEmployeeDetails(Employee employeeList[], int totalEmployees){
    for (int index = 0; index < totalEmployees; index++){
        cout << "\nEnter details for Employee " << index + 1 << ":\n";
        cout << "Employee Number: ";
        cin >> employeeList[index].employeeNumber;
        cin.ignore();
        cout << "Name: ";
        getline(cin, employeeList[index].employeeName);
        cout << "Department: ";
        getline(cin, employeeList[index].employeeDepartment);
        cout << "Salary: ";
        cin >> employeeList[index].employeeSalary;
    }
}

void displayEmployeeDetails(const Employee &employee){
    cout << "\n==== Employee Details =====\n";
    cout << "Employee Number : " << employee.employeeNumber << endl;
    cout << "Name : " << employee.employeeName << endl;
    cout << "Department : " << employee.employeeDepartment << endl;
    cout << "Salary : " << employee.employeeSalary << endl;
```

```
    cout << "=====\n";
}

void searchEmployeeByNumber(Employee employeeList[], int totalEmployees, int
searchEmployeeNumber){

    for (int index = 0; index < totalEmployees; index++){

        if (employeeList[index].employeeNumber == searchEmployeeNumber){

            displayEmployeeDetails(employeeList[index]);

            return;

        }

    }

    cout << "Employee with Employee Number " << searchEmployeeNumber << " not
found.\n";

}

int main(){

    const int TOTAL_EMPLOYEES = 10;

    Employee employeeList[TOTAL_EMPLOYEES];

    cout << "==== Employee Information System =====\n";

    inputEmployeeDetails(employeeList, TOTAL_EMPLOYEES);

    char userChoice;

    do{

        int searchEmployeeNumber;

        cout << "\nEnter Employee Number to search: ";

        cin >> searchEmployeeNumber;

        searchEmployeeByNumber(employeeList, TOTAL_EMPLOYEES,
searchEmployeeNumber);

        cout << "\nDo you want to search for another employee? (y/n): ";

        cin >> userChoice;

    } while (userChoice == 'y' || userChoice == 'Y');

    cout << "\nThank you for using the Employee Information System. Goodbye!\n";

    return 0;

}
```

Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\grv\Desktop\output> & .\project.exe
===== Employee Information System =====

Enter details for Employee 1:
Employee Number: 001
Name: Henry
Department: ext
Salary: 50

Enter details for Employee 2:
Employee Number: 011
Name: Jane
Department: exp
Salary: 20

Enter details for Employee 3:
Employee Number: 008
Name: eight
Department: exp
Salary: 30

Enter details for Employee 4:
Employee Number: 5090
Name: max
Department: st
Salary: 40999

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Enter details for Employee 5:
Employee Number: 5080
Name: holly
Department: st
Salary: 39999

Enter details for Employee 6:
Employee Number: 3050
Name: nancy
Department: st
Salary: 20999

Enter details for Employee 7:
Employee Number: 4090
Name: hopper
Department: army
Salary: 99999

Enter details for Employee 8:
Employee Number: 2050
Name: mike
Department: st
Salary: 2

Enter details for Employee 9:
Employee Number: 5050
Name: will
Department: st
Salary: 899

Enter details for Employee 10:
Employee Number: 4080
Name: steve
Department: st
Salary: 8999

Enter Employee Number to search: 5090

===== Employee Details =====
Employee Number : 5090
Name : max
Department : st
Salary : 40999
=====

Do you want to search for another employee? (y/n): y

Enter Employee Number to search: 5080

===== Employee Details =====
Employee Number : 5080
Name : holly
Department : st
Salary : 39999
=====

Do you want to search for another employee? (y/n): n

Thank you for using the Employee Information System. Goodbye!
PS C:\Users\grv\Desktop\output>
```

Program 18. : Program to accept and print a student's result using a structure having array inside it.

Soln:

```
#include <iostream>

#include <string>

using namespace std;

struct Student{

    string name;

    int rollNo;

    float marks[5];

    float totalMarks;

    float percentage;

};

void calculateResult(Student &student){

    student.totalMarks = 0;

    for (int i = 0; i < 5; i++){

        student.totalMarks += student.marks[i];

    }

    student.percentage = (student.totalMarks / 500) * 100;

}

int main(){

    Student student;

    cout << "Enter student name: ";

    getline(cin, student.name);

    cout << "Enter student roll number: ";

    cin >> student.rollNo;

    cout << "Enter marks for 5 subjects:\n";

    for (int i = 0; i < 5; i++)

    {

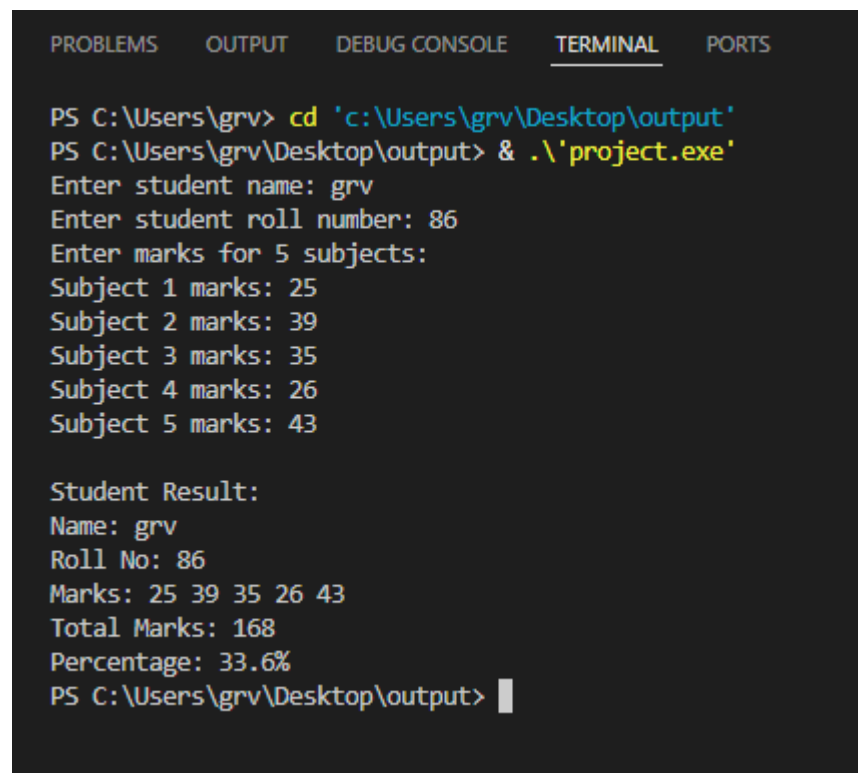
        cout << "Subject " << i + 1 << " marks: ";

        cin >> student.marks[i];

    }

}
```

```
calculateResult(student);  
  
cout << "\nStudent Result:\n";  
  
cout << "Name: " << student.name << endl;  
  
cout << "Roll No: " << student.rollNo << endl;  
  
cout << "Marks: ";  
  
for (int i = 0; i < 5; i++)  
{  
    cout << student.marks[i] << " ";  
}  
  
cout << endl;  
  
cout << "Total Marks: " << student.totalMarks << endl;  
  
cout << "Percentage: " << student.percentage << "%" << endl;  
  
return 0;  
}
```

Output:

The screenshot shows a terminal window with the following content:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
  
PS C:\Users\grv> cd 'c:\Users\grv\Desktop\output'  
PS C:\Users\grv\Desktop\output> & .\project.exe  
Enter student name: grv  
Enter student roll number: 86  
Enter marks for 5 subjects:  
Subject 1 marks: 25  
Subject 2 marks: 39  
Subject 3 marks: 35  
Subject 4 marks: 26  
Subject 5 marks: 43  
  
Student Result:  
Name: grv  
Roll No: 86  
Marks: 25 39 35 26 43  
Total Marks: 168  
Percentage: 33.6%  
PS C:\Users\grv\Desktop\output> |
```

Program 19. : Program to illustrate passing of structures by value.

Soln:

```
#include <iostream>

#include <string>

using namespace std;

struct Student{

    string name;

    int age;

    float marks;

};

void displayStudentDetails(Student student){

    cout << "\nInside the function (After passing by value):\n";

    cout << "Name: " << student.name << endl;

    cout << "Age: " << student.age << endl;

    cout << "Marks: " << student.marks << endl;

    student.marks += 5;

    cout << "\nModified Marks Inside Function: " << student.marks << endl;

}

int main(){

    Student student1;

    cout << "Enter student name: ";

    getline(cin, student1.name);

    cout << "Enter student age: ";

    cin >> student1.age;

    cout << "Enter student marks: ";

    cin >> student1.marks;

    cout << "\nBefore passing to function (By value):\n";

    cout << "Name: " << student1.name << endl;

    cout << "Age: " << student1.age << endl;

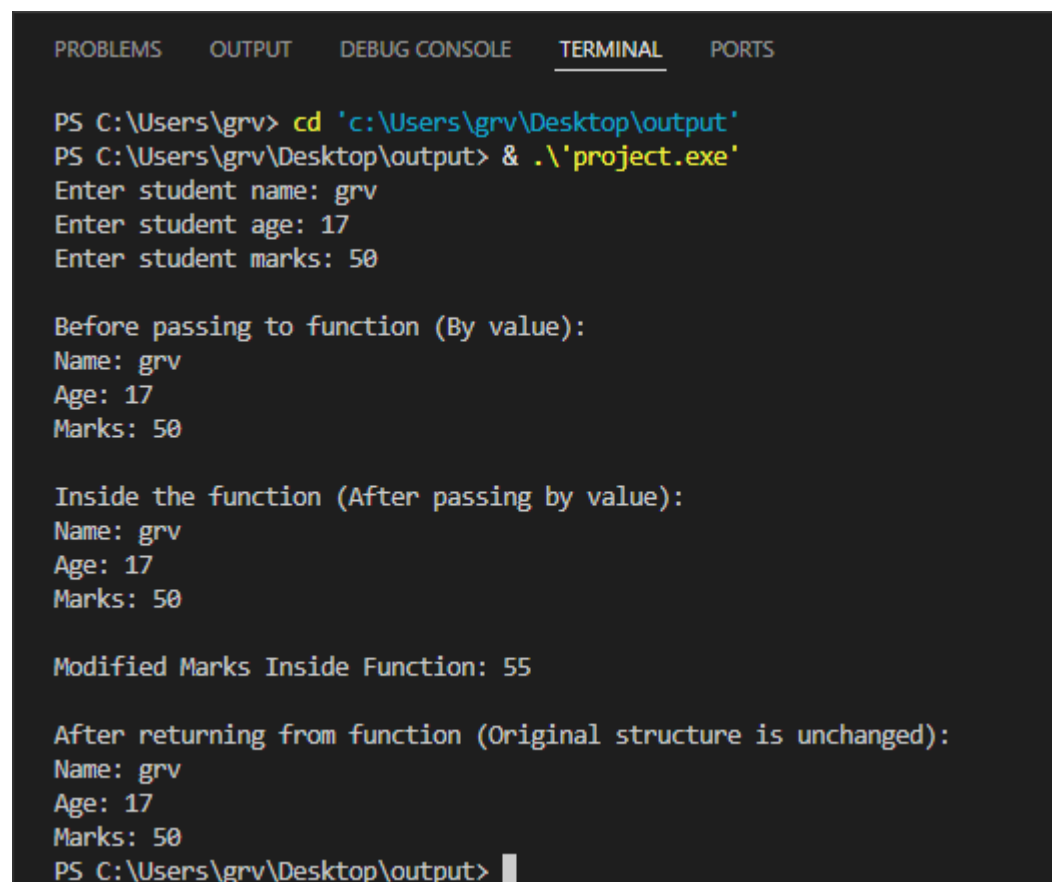
    cout << "Marks: " << student1.marks << endl;

    displayStudentDetails(student1);

}
```

```
cout << "\nAfter returning from function (Original structure is unchanged):\n";  
cout << "Name: " << student1.name << endl;  
cout << "Age: " << student1.age << endl;  
cout << "Marks: " << student1.marks << endl;  
return 0;  
}
```

Output:



The screenshot shows a terminal window with the following output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
  
PS C:\Users\grv> cd 'c:\Users\grv\Desktop\output'  
PS C:\Users\grv\Desktop\output> & .\project.exe  
Enter student name: grv  
Enter student age: 17  
Enter student marks: 50  
  
Before passing to function (By value):  
Name: grv  
Age: 17  
Marks: 50  
  
Inside the function (After passing by value):  
Name: grv  
Age: 17  
Marks: 50  
  
Modified Marks Inside Function: 55  
  
After returning from function (Original structure is unchanged):  
Name: grv  
Age: 17  
Marks: 50  
PS C:\Users\grv\Desktop\output>
```

Program 20. : Program to illustrate passing of a structure by reference.

Soln:

```
#include <iostream>

#include <string>

using namespace std;

struct Student{

    string name;

    int age;

    float marks;

};

void modifyStudentDetails(Student &student){

    cout << "\nInside the function (After passing by reference):\n";

    cout << "Name: " << student.name << endl;

    cout << "Age: " << student.age << endl;

    cout << "Marks: " << student.marks << endl;

    student.marks += 10;

    cout << "\nModified Marks Inside Function: " << student.marks << endl;

}

int main(){

    Student student1;

    cout << "Enter student name: ";

    getline(cin, student1.name);

    cout << "Enter student age: ";

    cin >> student1.age;

    cout << "Enter student marks: ";

    cin >> student1.marks;

    cout << "\nBefore passing to function (By reference):\n";

    cout << "Name: " << student1.name << endl;

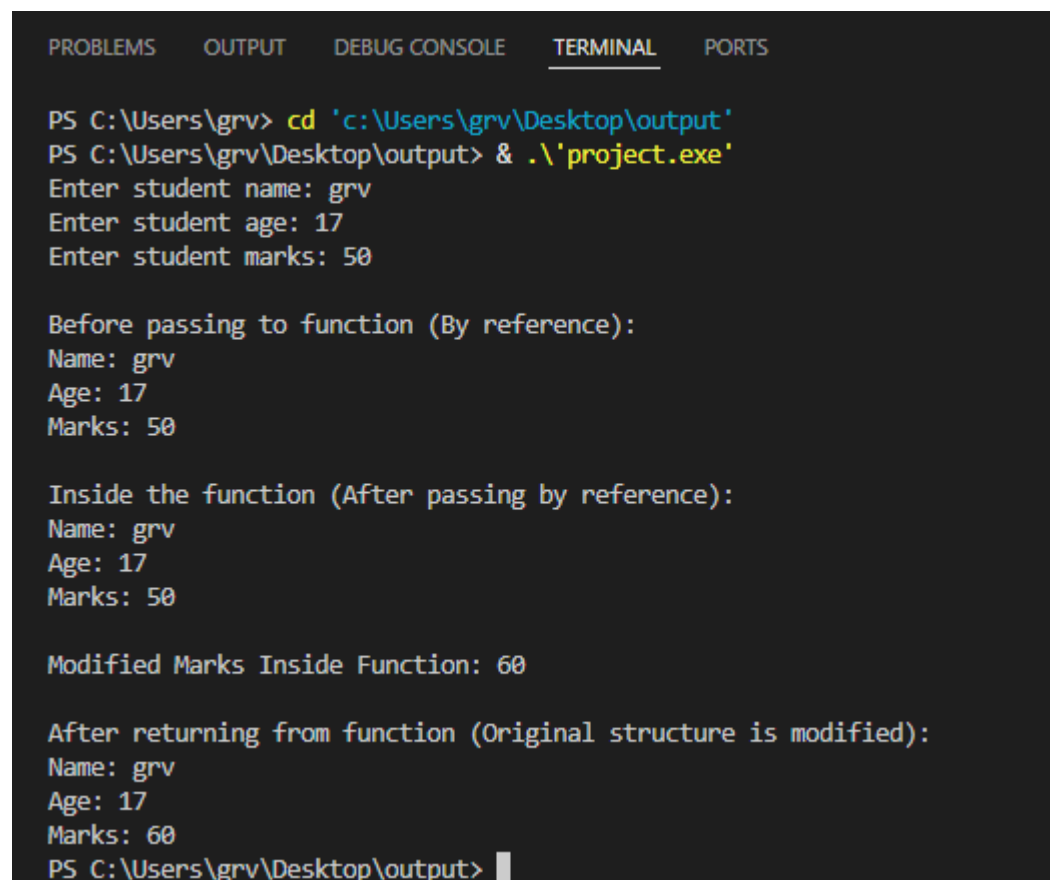
    cout << "Age: " << student1.age << endl;

    cout << "Marks: " << student1.marks << endl;

    modifyStudentDetails(student1);
```

```
cout << "\nAfter returning from function (Original structure is modified):\n";  
cout << "Name: " << student1.name << endl;  
cout << "Age: " << student1.age << endl;  
cout << "Marks: " << student1.marks << endl;  
return 0;  
}
```

Output:



The screenshot shows a terminal window with the following output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
  
PS C:\Users\grv> cd 'c:\Users\grv\Desktop\output'  
PS C:\Users\grv\Desktop\output> & .\'project.exe'  
Enter student name: grv  
Enter student age: 17  
Enter student marks: 50  
  
Before passing to function (By reference):  
Name: grv  
Age: 17  
Marks: 50  
  
Inside the function (After passing by reference):  
Name: grv  
Age: 17  
Marks: 50  
  
Modified Marks Inside Function: 60  
  
After returning from function (Original structure is modified):  
Name: grv  
Age: 17  
Marks: 60  
PS C:\Users\grv\Desktop\output>
```